Sequences — nth terms · The difference method · Arithmetic sequences and series

Calculator allowed. Marks in brackets. ★ = harder question. $S_n = \frac{n}{2}\left[2a + (n-1)d\right]$.

1 Write down the first three terms of the sequence whose nth term is [3]

a $5n + 2$ **b** $n^2 - n$ **c** $\dfrac{2n+1}{n+2}$

> Substitute $n = 1, 2, 3$. **a** $7,\ 12,\ 17$. **b** $0,\ 2,\ 6$. **c** $1,\ \frac{5}{4},\ \frac{7}{5}$.

2 Use the difference method to find the next three terms of each sequence. [3]

a $3,\ 7,\ 15,\ 27,\ 43,\ \ldots$ **b** $20,\ 17,\ 14,\ 11,\ \ldots$ **c** $1,\ 2,\ 4,\ 7,\ 11,\ \ldots$

> **a** Differences $4, 8, 12, 16$ — they go up by 4, so the next differences are $20, 24, 28$: $63,\ 87,\ 115$. **b** Constant difference -3: $8,\ 5,\ 2$. **c** Differences $1, 2, 3, 4$, next $5, 6, 7$: $16,\ 22,\ 29$.

3 Find a formula for the nth term of each sequence. [4]

a $6,\ 10,\ 14,\ 18,\ \ldots$ **b** $93,\ 86,\ 79,\ 72,\ \ldots$

> **a** The difference is constant, 4, so the formula is $4n + b$. When $n = 1$ it must equal 6: $4 + b = 6 \Rightarrow b = 2$. nth term $= 4n + 2$.
>
> **b** Constant difference -7, so $-7n + b$; $n = 1$: $-7 + b = 93 \Rightarrow b = 100$. nth term $= 100 - 7n$.

4 The 3rd term of an arithmetic sequence is 11 and the 8th term is 31. Find the first term a and the common difference d. [4]

> nth term $= a + (n-1)d$. So $a + 2d = 11$ and $a + 7d = 31$. Subtracting: $5d = 20 \Rightarrow d = 4$. Then $a = 11 - 8 = 3$.

5 The 6th term of an arithmetic sequence is 23 and the 15th term is 50. Find the sum of the first 30 terms. [5]

> $a + 5d = 23$ and $a + 14d = 50$. Subtracting: $9d = 27 \Rightarrow d = 3$, so $a = 23 - 15 = 8$.
>
> $S_{30} = \dfrac{30}{2}\left[2 \times 8 + 29 \times 3\right] = 15 \times (16 + 87) = 15 \times 103 = 1545$.

6 The sum of the first 24 terms of an arithmetic sequence is 1080. The common difference is 2. Find the first term. [3]

> $S_{24} = \dfrac{24}{2}\left[2a + 23 \times 2\right] = 12\,(2a + 46) = 1080 \Rightarrow 2a + 46 = 90 \Rightarrow a = 22$.

7 Logs are stacked in rows. The bottom row has 30 logs, each row above has one log fewer than the row below, and the top row has 6 logs. How many logs are in the stack? [3]

> The rows form an arithmetic sequence $30, 29, \ldots, 6$: $a = 30$, $d = -1$. Number of rows: $30 - 6 + 1 = 25$. Sum $= \frac{25}{2}(30 + 6) = 25 \times 18 = 450$ logs. (Using the first-and-last form $S_n = \frac{n}{2}(a + l)$; the formula with d gives the same: $\frac{25}{2}[60 - 24] = 450$.)

Total: 25 marks

Check: a constant first difference a means the nth term is $an + b$ — find b from the first term · arithmetic sequence: nth term $a + (n-1)d$ · two given terms → two equations, subtract to find d · $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$ · count the number of terms carefully (last − first + 1 when $d = \pm 1$).